

Practice 12 R: ARFIMA Model

First of all we can simulate an ARFIMA model:

- `n<-1000`
- `x<-arima.sim(n, model=list(phi=c(0.2,0.1),`
- `dfrac=0.4,theta=0.9))`
- `plot(x, type="l")`

Now fit the best model:

- `install.packages("fracdiff")`
- `library(fracdiff)`
- `fit1<-fracdiff(x, nar=2, nma=1, drange=c(0,0.5))`
- `summary(fit1)`
- `d<-fit1$d`
- `fit2=diffseries(x,d)`
- `acf(fit2)`
- `fit3<-arima(fit2,order=c(2,0,1))`
- `fit4<-resid(fit3)`
- `acf(fit4)`
- `Box.test(fit4, lag=log(n), type=c("Ljung-Box"))`

Note that fit4 is an IID noise:

But in fact, note that the estimation is not good, the estimated d is far from 0.4.

Change different values for d, phi, theta, n....

Another example:

- `library(fracdiff)`
- `n<-1000`
- `sample<-fracdiff.sim(n, ar=0.9, d=0.4)`
- `x<-sample$series`
- `plot(x,type="l")`
- `fit1<-fracdiff(x,nar=1)`
- `summary(fit1)`
- `d<-fit1$d`
- `y1<-diffseries(x,d)`
- `acf(y1)`
- `y2<-arima(y1,order=c(1,0,0))`
- `z<-resid(y2)`
- `acf(z)`
- `Box.test(z,log(n),type=c("Ljung-Box"))`

For (1,d,0) models, this code is quite good, but if we change to an (0,d,1) things become worse. Try it.

Consider now the real series of quarterly data of Australian residents from 1972 to 1993 (88 data).

- `library("datasets")`
- `x<-austres`
- `acf(x)`
- `dx<-diff(x)`
- `acf(dx)`
- `fdx<-fracdiff(dx,nar=0,nma=0, drange=c(0,0.5))`
- `summary(fdx)`
- `d=fdx$d`

- d
- resid<-diffseries(dx,d)
- y<-resid-mean(resid)
- acf(y)
- Box.test(y,lag=log(88),type=c("Ljung-Box"))

Practice 14 R: GARCH models

1. White but not IID noise residuals.

Let us analyze the daily returns of the S&P500:

- library(MASS)
- data(SP500)
- plot(SP500, type="l")
- acf(SP500)
- Box.test(SP500, type="Ljung-Box")

Clearly we have a white noise.

We can compute the mean:

- m<-mean(SP500)
- x<-SP500-m
- plot(x, type="l")
- acf(x)

Of course, we are under the same situation. Note that the series x is not stationary in variance.

But do the following:

- y<-x^2
- plot(y, type="l")
- acf(y)
- Box.test(y, type="Ljung-Box")

What is happening? Note that x is a white noise but y not.

We need a new theory to give structure to this type of processes.

2. Generating ARCH and GARCH models.

- a0<-0.25
- a1<-0.5
- x<-numeric(9999)
- x0<-rnorm(1)
- for (i in 1:9999){
- x0<-rnorm(1,sd=sqrt(a0+a1*x0^2))
- x[i]<-x0
- }
- plot(x, type="l")

An alternative system for a GARCH(1,1) is

- n<-10000
- a0<-0.1
- a1<-0.4
- b1<-0.2
- w<-rnorm(n)
- y<-rep(0,n)
- h<-rep(0,n)
- for(i in 2:n){
- h[i]<-a0+a1*(y[i-1]^2)+b1*h[i-1]
- y[i]<-w[i]*sqrt(h[i])
- }
- plot(y, type="l")
- acf(y)
- acf(y^2)

3. Do the same with an ARCH(2).

4. Fitting GARCH models

Now we adjust a GARCH model to SP500 data.

- `library(MASS)`
- `library(tseries)`
- `data(SP500)`
- `plot(SP500, type="l")`
- `x<-SP500-mean(SP500)`
- `plot(x,type="l")`
- `sp<-garch(x, trace=F)`
- `res<-sp$res[-1]`
- `plot(res,type="l")`
- `acf(res)`
- `acf(res^2)`
- `confint(sp)`
- `garch(x,order=c(1,1), trace=F)`

The instruction “confint” gives the confidence intervals of the parameters of the estimated GARCH model. The last instruction gives the estimated parameters.